AgriAdvise Mobile Frontend API Handoff

Backend contract and user-flow guide for the mobile application

Base URL: `https://<host>/api/v1`. Send `Authorization: Bearer <token>` for authenticated endpoints. Use JSON unless an endpoint explicitly accepts a multipart media upload.

Download this document: `GET /api/v1/documentation/mobile-frontend-handoff`. The response is a PDF attachment.

> **Source of truth:** The backend selects agricultural advice. The app only collects context and presents the recommendation, action steps, warnings and contacts it receives. Do not place API keys, Tranzak credentials, or farmer telephone numbers in the mobile app.

1. Farmer mobile journey

1. **Account:** register or sign in, then securely save the returned access token.
2. **Profile:** load the farmer profile and save language/knowledge level. Use this to choose simple or detailed presentation.
3. **Guided help:** start a questionnaire with region and locality, then securely retain the returned `conversation_id`. Display one question at a time.
4. **Every question:** show the supplied options *and* a text box. Include “I do not know”. The farmer may choose an option or type a different issue.
5. **Adapt:** submit the selected value or free text with `conversation_id`. The backend records the message and understands it as a new question or an answer to the pending prompt.
6. **Recommendation:** when enough context has been supplied, `POST /questionnaire/answer` returns `complete: true` with the recommendation. Stop asking questions and render it.
7. **Follow-up:** allow a new guided-help session without overwriting earlier conversations. Open one later with the conversation-history endpoint.

Location behavior: region/locality facts are checked before asking location-related questions. The app must not force a terrain question when the backend already knows the relevant local context. When information is absent, display the next backend question.

2. Recommended screen/API sequence

Screen	Request	Frontend responsibility
Welcome/sign in	`POST /auth/register` or `POST /auth/login`	Securely retain the token; recover gracefully from invalid credentials.
Profile	`GET /profile/me`, `PUT /profile/me`	Store and edit language/knowledge-level preferences.
Start help	`POST /questionnaire/start`	Send farmer ID, region, locality and profile level; save returned `conversation_id` and render `step`.
Question step	`POST /questionnaire/answer`	Send `conversation_id` and either an option or free text; render returned `step` or recommendation.
Conversation history	`GET /questionnaire/conversations/{conversation_id}`	Load the farmer-owned ordered history, collected context and final result when available.
Advice result	`POST /questionnaire/complete`	Optional explicit completion for a fully collected context; present returned recommendation exactly as supplied.
Optional direct advice	`POST /advisory/mobile`	For experienced users or a saved form; use the same validated context fields.

3. Voice and picture input

- **Voice-to-text:** upload `file` and optional `language` to `POST /advisory/transcribe`. Put returned `text` in the same free-text questionnaire input for the farmer to confirm/edit.
- **Voice answer:** `POST /advisory/voice` creates spoken advice where the configured provider is available.
- **Picture evidence:** upload `file`, `farmer_id`, optional `crop_id` and `request_id` to `POST /media/images`. Keep the returned `image_id` and include it in `evidence.image_references`.
- **Optional image assessment:** call `POST /image-analysis` with `image_id`, optional `crop_id`, and `analysis_type`. It is supporting evidence, never a substitute for farmer description or backend recommendation.

4. Complete endpoint inventory

Endpoint	Purpose for frontend

`POST /auth/register`
`POST /auth/login`
`GET /auth/me`
	Registration, login, current account.

`GET, PUT /profile/me`	Farmer profile and preferences.

`POST /questionnaire/start`
`POST /questionnaire/answer`
`POST /questionnaire/complete`
`GET /questionnaire/conversations/{conversation_id}`
	Adaptive guided conversations, automatic final advice, and saved farmer-owned conversation history.

`POST /advisory/mobile`
`POST /advisory/voice`
`POST /advisory/transcribe`
	Direct recommendation, spoken result, and voice-to-text.

`POST /media/images`
`GET /media/images/{image_id}`
`GET /media/images/{image_id}/metadata`
`DELETE /media/images/{image_id}`
	Farmer-owned picture upload, display, metadata and deletion.

`POST /image-analysis` `GET /image-analysis/references`	Optional image evidence assessment and supported reference listing.
`GET /recommendations/{recommendation_id}`	Saved recommendation and its trace.
`GET /crops` `GET /crops/{crop_id}`	Crop catalogue/profile.

`GET /education/crops`
`GET /education/crops/{crop_id}`
`POST /education/learn/curriculum`
	Learning content and learning plan.

`GET /contacts/{contact_type}`	Verified specialist/vendor cards when returned in a recommendation.

`POST /technical-fiches`
`GET /technical-fiches`
`GET /technical-fiches/{fiche_id}`
`PUT /technical-fiches/{fiche_id}/status`
`GET /technical-fiches/{fiche_id}/download`
	Create, list, view, update status and download technical fiches.

`GET /privacy/me`
`GET /privacy/export`
`DELETE /privacy/me`
	Privacy status, personal-data export and account removal.

`GET /health`
`GET /ready`
`GET /knowledge/version`
	Operational checks; usually used by support/release tooling, not normal farmer screens.

`POST /advisory/sms`
`POST /sms/simulate`
`GET /sms/inbox/{recipient_id}`
	Legacy SMS-advice flow and safe simulator testing. Not a place for the mobile app to store recipients.

`POST, GET /notifications/subscriptions` `POST /notifications/trigger/{recipient_id}`	Legacy notification sandbox.

5. Admin and operations endpoints

These require an admin token and belong to the administration dashboard, not normal farmer navigation.

Endpoint	Purpose

`GET, POST /admin/contacts`
`POST /admin/contacts/{contact_id}/verification`
`POST /admin/contacts/{contact_id}/activation`
	Manage and approve specialists/vendors.

`GET /admin/overview` `GET /admin/providers`	Operational overview and provider status.
`GET /admin/knowledge` `POST /admin/knowledge/import`	Inspect/import validated knowledge.
`POST /admin/retention/run`	Run configured data-retention work.
`GET /dashboard` `GET /dashboard/sms-receiver`	Built-in browser operations pages; they are outside the API prefix.

Regional SMS administration

These administrator-only endpoints manage phone number plus Cameroon region only; they do not store crop, passport, or recommendation context.

1. `GET, POST /admin/sms/registrations`: list or register recipients.
2. `POST /admin/sms/broadcasts/preview`: show the exact active recipient count before confirmation.
3. `POST /admin/sms/broadcasts/send`: after confirmation, re-resolve active recipients, send, and return `sent`, `failed`, and `recipient_count`.
4. `GET /admin/sms/deliveries?limit=100`: display SMS history.

Use `{"kind":"region","region":"South"}` for a regional message or `{"kind":"cameroon"}` for a Cameroon-wide message. Automatic SMS is separate from normal recommendations: only validated, explicitly scoped alerts are eligible, and the worker remains idle without them.

6. Error and security behavior

- `400`/`422`: show an input-specific validation message and keep the entered form data.
- `401`: refresh/re-authenticate, then retry only after the user signs in.
- `403`: show that the content is not owned by the current farmer or requires a higher role.
- `404`: show unavailable content; do not guess identifiers or agricultural content.
- Never log bearer tokens, voice files, image files, farmer phone numbers or API credentials in the mobile client.

Developer reference: the live OpenAPI interface is available at `https://<host>/docs` and is authoritative for field-level schemas.